



Securing Apache Airflow with Keycloak

A Deep Dive into the Keycloak Auth Manager



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About us



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Agenda

01 Auth managers at a glance

What Airflow hands off, and why Keycloak is a good fit

02 Inside the Keycloak auth manager

Logging in, checking permissions, and setting it all up

03 Demos

A single team, then several teams on one Airflow



PART ONE

Auth managers at a glance

What is an auth manager?

The single component Airflow asks "who is this?" and "may they do this?"

Authentication

- Logs the user in and out
- Issues the Airflow JWT
- Refreshes expiring tokens

Authorization

- `is_authorized_*` per resource
- Filters team-centric resources: Dags, Connections, Variables, Pools
- Filters the menu items

Where the auth manager sits



← Allowed: the API runs the action and returns **200**. Denied: **403**, and Airflow never touched the Dag.

Why Keycloak?

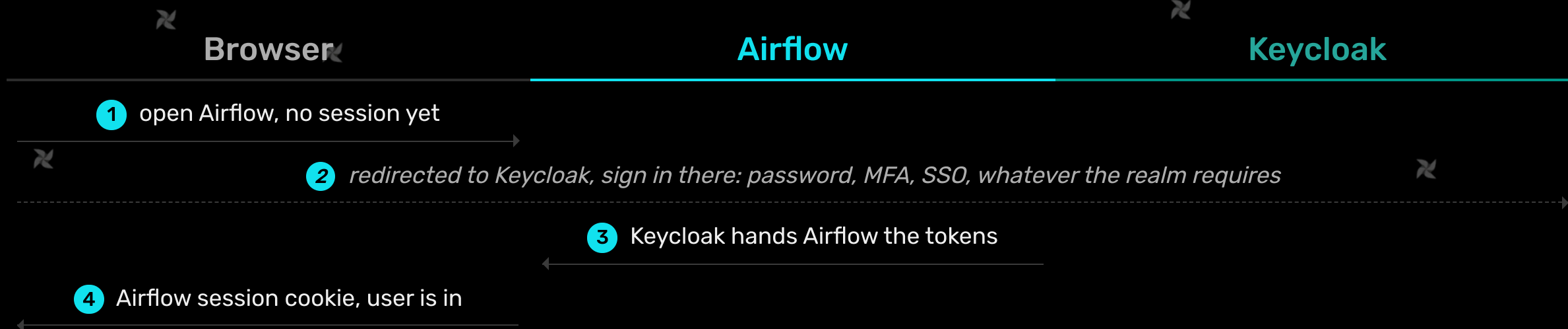
- ✓ Open source, self-hosted, widely deployed
- ✓ Standard OpenID Connect, so SSO and MFA come for free
- ✓ Fine-grained authorization built in



PART TWO

How does the Keycloak auth manager work?

Authentication: standard OIDC



Authorization: every check is a token request

Airflow does not decide. It asks Keycloak for a UMA ticket and reads the HTTP status.

```
POST /realms/{realm}/protocol/openid-connect/token
Authorization: Bearer <user access token>

grant_type    = urn:ietf:params:oauth:grant-type:uma-ticket
audience     = {client_id}
permission    = Dag#GET                # <resource>#<scope>
claim_token   = base64({"resource_id": ["my_dag"], "dag_entity": ["RUN"]})
```

Keycloak says	Airflow concludes
200	Authorized
401 / 403	Denied
400	Misconfiguration → raises (except a missing team resource, which is treated as a deny)

Resources & scopes

Resources

One Keycloak resource per Airflow concept, plus one per menu item.

Dag Connection Variable Pool Asset AssetAlias
Backfill Configuration Team View Menu Custom

Scopes

The HTTP verbs, plus two synthetic ones.

GET POST PUT DELETE MENU LIST

LIST replaces GET when no specific resource id is in play – "can you see the collection at all?"

Setting it up

1. Point Airflow at Keycloak

```
[core]
auth_manager = airflow.providers.keycloak.auth_manager.keycloak_auth_manager.KeycloakAuthManager

[keycloak_auth_manager]
client_id = airflow
client_secret = ...
realm = airflow
server_url = http://localhost:48080
requests_pool_size = 10
requests_retries = 3
```

2. Seed the Keycloak client

```
# scopes + resources + permissions in one go
airflow keycloak-auth-manager create-all --username admin --password admin

# see what it would do first
airflow keycloak-auth-manager create-all --username admin --dry-run
```

Also available individually: create-scopes, create-resources, create-permissions.



PART THREE

Demo


```
kc ➤ for r in Admin Op User Viewer SuperAdmin; do kcadm.sh create roles -r airflow -s name=$r; done
```

I

```
af ➤
```

localhost:28080/auth/login? ...

Not Secure http://keycloak:48080/admin/master/console/#/airflow/users/57cdb1a6-2138-48a8-a9e7-7a98abfddfdde/role-mapping

200%

VPN

KEYCLOAK

admin

airflow

Manage

Clients

Client scopes

Realm roles

Users

Groups

Sessions

Events

Configure

Realm settings

Authentication

Identity providers

Users > User details

viewer

Enabled

Action

Details

Attributes

Credentials

Role mapping

Groups

Consents

Identity provider links

Sessions

Search by name

→

☒ Hide inherited roles

Assign role

Unassign

1 - 2

<

>

☐	Name	Inherited	Description	
☐	default-roles-airflow	False	\${role_default-roles}	
☐	Viewer	False	—	

1 - 2<>

Sign in to airflow

Keycloak Administration

Not Securehttp://keycloak:48080/admin/master/console/#/airflow/users/058f3d89-f30c-4d11-ae61-27dc1e061bb2/role-mappingVPNMM

KEYCLOAK

airflow

Manage

Clients

Client scopes

Realm roles

Users

Groups

Sessions

Events

Configure

Realm settings

Authentication

Identity providers

User federation

Users > User details

vinnie

Enabled

Action

DetailsAttributesCredentialsRole mappingGroupsConsentsIdentity provider linksSessions

Search by name→☒ Hide inherited rolesAssign roleUnassign

☐ Name	Inherited	Description	
☐ default-roles-airflow	False	\${role_default-roles}	
☐ Viewer	False	-	

1-2<>

Multi-team: resources get a team suffix

With `core.multi_team = True`, team-scoped resources become `<Resource>:<team>`:

`Dag:team-a` `Connection:team-a`

`Variable:team-a` `Pool:team-a` `Team:team-a`

Access is then an **aggregate policy**: **group membership** **AND** **role**. Being in group `team-a` with role `Op` grants `Op-team-a` – and nothing in team B.

```
# one command per team
airflow keycloak-auth-manager create-team \
    team-a --username admin

# then put people in it
airflow keycloak-auth-manager add-user-to-team \
    alice team-a --username admin
```

Keep them in sync

Add a team in Airflow without running `create-team` and Keycloak answers `invalid_resource`. The auth manager degrades that to a deny, so `/dags` still renders for the other teams.

Sign in to airflow

Keycloak Administration

Not Securehttp://keycloak:48080/admin/master/console/#/airflow/clients/6c4fc54e-a141-433c-beee-ea4dc257d161/authorization/policies

200%VPN

KEYCLOAK

admin

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Manage

Clients

Client scopes

Realm roles

Users

Groups

Sessions

Events

Configure

Realm settings

Authentication

Identity providers

Settings

Resources

Scopes

Policies

Permissions

Evaluate

Export

Search for permission

Create client policy

1 - 10

Name	Type	Dependent permission	Description
> Allow-Admin	Role	Allow-Admin-team-a 2 more	
> Allow-Admin-team-a	Aggregate	ReadOnly-team-a 4 more	
> Allow-Admin-team-b	Aggregate	ReadOnly-team-b 4 more	
> Allow-Op	Role	Allow-Op-team-a 2 more	
> Allow-Op-team-a	Aggregate	ReadOnly-team-a 3 more	
> Allow-Op-team-b	Aggregate	ReadOnly-team-b 3 more	
> Allow-SuperAdmin	Role	GlobalList 3 more	
> Allow-Team-team-a	Group	Allow-Viewer-team-a 3 more	
> Allow-Team-team-b	Group	Allow-Viewer-team-b 3 more	

Sign in to airflow

Keycloak Administration

Not Secure http://keycloak:48080/admin/master/console/#/airflow/users/9b4a99da-f2fe-430e-b692-b430b12311c9/groups

200%

VPN

admin

KEYCLOAK

airflow

Manage

Clients

Client scopes

Realm roles

Users

Groups

Sessions

Events

Configure

Realm settings

Authentication

Identity providers

Users > User details

user-b

Enabled

Action

Details

Attributes

Credentials

Role mapping

Groups

Consents

Identity provider links

Sessions

Search group

Join Group

Direct membership

Leave

Who will appear in this group list?

1-1

Group membership

Path

team-b

/team-b

Leave

1-1

Takeaways

- Auth managers make identity a **pluggable** concern in Airflow 3
- The Keycloak auth manager gives you real **SSO** plus **per-DAG authorization**, managed entirely outside Airflow
- Multi-team isolation is **group + role** — onboarding a team is **one command**: `create-team`
- All of it runs on **standard Keycloak features** — resources, scopes, groups, roles, aggregate policies — curated by the open source community



Questions?



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